batch norm =

$$\mu = \frac{1}{m}\sum x_i$$

$$\sigma^2 = \frac{1}{m-1}\sum (x_1 - \mu)^2$$

$$\boxed{\hat{x}_1 = \frac{x_i - \mu}{\sqrt{\sigma^2 + \varepsilon}}}$$

$$y_i = \gamma \hat{x}_i + \beta$$

$$\frac{d}{dx}\mu = \frac{d}{dx_i}\left(\frac{1}{m}\sum x_i\right)$$

$$= \frac{1}{m}\frac{d}{dx_i}\Big(\underset{\substack{\downarrow \\ 1}}{x_i} + \overbrace{\ldots\ldots x_m + \ldots x_0}^{0}\Big\}$$

$$= \frac{1}{m} \times 1 = \frac{1}{m}$$

$$\frac{d}{dx}\sigma^2 = \frac{d}{dx}\left(\frac{1}{m-1}\sum (x_1 - \mu)^2\right)$$

$$= \frac{1}{m-1}\sum_j \frac{d}{dx_i}(x_j - \mu)^2$$

$$= \frac{1}{m-1}\sum_j \left(\frac{d}{dx_i}(x_j - \mu)^2\right)$$

$$= \frac{1}{m-1}\sum_j 2(x_j - \mu)\frac{d}{dx_i}(x_j - \mu)$$

if $i=j$:- $\frac{d}{dx_i} = 1 - \frac{1}{m}$ else: $-\frac{1}{m}$

$$= \frac{2}{m-1}\left[(x_i - \mu)\left(1 - \frac{1}{m}\right) + \sum_{i \neq j}(x_j - \mu)\left(-\frac{1}{m}\right)\right.$$

$$= \frac{2}{m-1}\left[\left(1 - \frac{1}{m}\right)(x_i - \mu) + \underbrace{\frac{1}{m}(x_1 - \mu)}\right]$$

as $\sum_j (x_j - \mu) = 0$

$\therefore$ for $i \neq j \to -(x_1 - \mu)$

$$= \frac{2(x_1 - \mu)}{m-1}$$

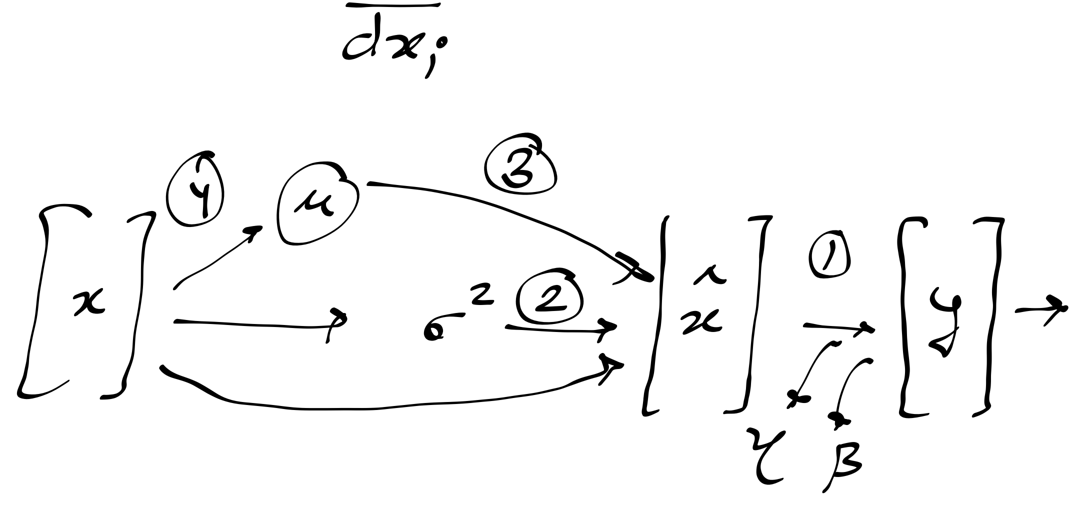
$$\frac{d}{dx_i}\hat{x}_i = \frac{d}{dx_i}\frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

$$= \frac{1}{\sqrt{\sigma^2 + \epsilon}} \cdot 1$$

$$= (\sigma^2 + \epsilon)^{-1/2}$$

Have:- $\frac{dL}{dy_i}$

Want :- $\frac{dL}{dx_i}$



① $y_i = \gamma \hat{x}_i + \beta$

$$\boxed{\frac{dy_i}{dx_i} = \gamma}$$

$$\frac{dL}{d\hat{x}_1} = \frac{dl}{dy_i} \times \frac{dy_i}{d\hat{x}_i}$$

$$= \frac{dl}{dy_i}\gamma$$

② $\frac{dl}{d\sigma^2} = \sum_i \frac{dl}{d\hat{x}_i}\frac{d\hat{x}_i}{d\sigma^2}$ ← as we have multiple expr (i...)

$$= \gamma \sum_i \frac{d}{d\sigma^2}\left[(x_i - \mu)(\sigma^2 + \epsilon)^{-1/2}\right] \cdot \frac{dl}{dy_i}$$

$$= -\frac{1}{2}\gamma \sum_i \frac{dl}{dy_i} \cdot (x_i - \mu)(\sigma^2 - \epsilon)^{-3/2}$$

③ $\frac{dl}{d\mu} = \underbrace{\left(\sum_i \frac{dl}{d\hat{x}_i}\frac{d\hat{x}_i}{d\mu}\right)} + \frac{dL}{d\sigma^2} \cdot \underbrace{\frac{d\sigma^2}{d\mu}}$

As from μ, we derive multiple $\hat{x}$ & one scalar σ^2

$$= \gamma \sum_i \frac{d}{d\mu}\left(\frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}}\right)\frac{dL}{dy_i}$$

$$= -\gamma \sum_i \frac{dl}{dy_i} \cdot (\sigma^2 + \epsilon)^{-1/2}$$

$$= \frac{d}{d\mu}\left[\frac{1}{m-1}\sum_i (x_i - \mu)^2\right]$$

$$= \frac{-2}{m-1}\sum_i (x_i - \mu)$$

$$= \frac{-2}{m-1}\left(\sum_i x_i - \sum_i \mu\right)$$

$$= \frac{-2}{m-1}(m\mu - m\mu)$$

$m\mu \to \mu = \frac{1}{m}\sum x_i$ ← uniform dist.

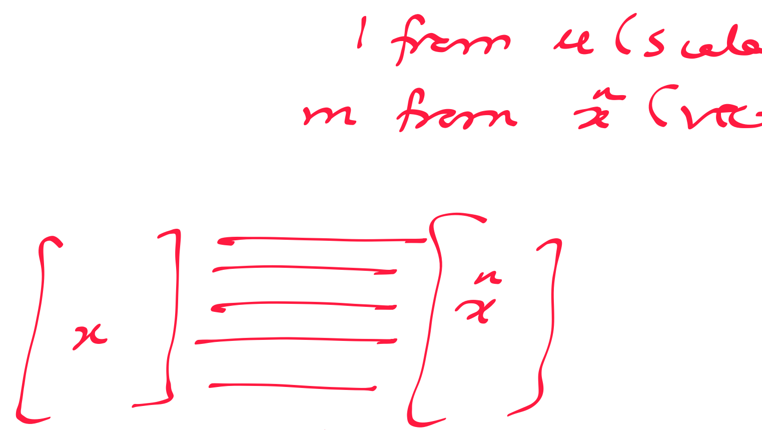
$$= 0.$$

$$\therefore \frac{dl}{d\mu} = -\gamma \sum_i \frac{dl}{dy_i}(\sigma^2 + \epsilon)^{-1/2}$$

3. $\frac{dl}{dx_i} = \frac{dl}{d\mu}\frac{d\mu}{dx_i} + \frac{dL}{d\sigma^2}\frac{d\sigma^2}{dx_i}$

$$+ \frac{dl}{d\hat{x}_i}\frac{d\hat{x}_i}{dx_i}$$

back prop:- 1 from σ^2 (scalar)
1 from μ (scalar)
m from $\hat{x}$ (vector)



we have all of these, let's plug in.

$$= -\gamma \sum_j \frac{dl}{dy_j}(\sigma^2 + \epsilon)^{-1/2} \times \frac{1}{m} +$$

$$-\frac{1}{2}\gamma \sum_j \frac{dL}{dy_j}(x_j - \mu)(\sigma^2 + \epsilon)^{-3/2} \times \frac{2}{m-1}(x_i - \mu) +$$

$$\gamma \frac{dL}{dy_i} \times (\sigma^2 + \epsilon)^{-1/2}$$

$$= \frac{\gamma(\sigma^2 + \epsilon)^{-1/2}}{m}\left[m\frac{dl}{dy_i} - \sum_j \frac{dl}{dy_j} - \frac{m}{m-1}\hat{x}_i \sum_j \frac{dL}{dy_j}\hat{x}_j\right]$$